Quantum Mechanics Tutorial-II
CY1001
Jan-May 2026

1. Prove that the kinetic energy operator, $\hat{T}$, is Hermitian.

2. Wavefunctions of a particle constrained to move on a circular ring of radius *r* are, $\psi_m(\theta) = \frac{1}{\sqrt{2\pi}} e^{im\theta}$ where $m = 0, \pm1, \pm2, \pm3$ ...
   Prove that the wavefunctions for this particle form an orthonormal set.

3. According to quantum mechanics, the angular momentum of a rotating body takes up arbitrary values (True/False)

4. What is the longest wavelength of light absorbed by a conjugated linear polyene containing 7 double bonds? Assume that the average length of the C-C bond is 140 pm. Also assume that no more than two electrons can be present in a given level and use a particle in a 1-D box model, ignoring the end effects.

5. The four π electrons on cyclobutadiene may be approximated as moving in a two-dimensional square box. Using the average C-C bond length of 1.4 Angstroms, calculate the energy of transition from the highest occupied state to the lowest unoccupied state, assuming that each state can have a maximum of two electrons.

6. The size of a quantum dot which consists of 10 atoms is increased to 10^5 atoms and subsequently the absorption wavelength of the quantum dot is changed from λ_1 and λ_2. The relation between λ_1 and λ_2 is always $\lambda_1 < \lambda_2$. (True/False)

7. In spherical polar coordinate system, the polar and azimuthal angles (θ and φ) have the boundary conditions-

   a. θ = 0 to π and Φ = 0 to π
   b. θ = 0 to 2π and Φ = 0 to π
   c. θ = 0 to 2π and Φ = 0 to 2π
   d. θ = 0 to π and Φ = 0 to 2π

8. Identify the correct statement(s)
   (a) The total number of nodes in a hydrogen atom orbital is $n - 1$, where n is the principal quantum number.
   (b) There are two types of nodes in the hydrogen atom: radial nodes and angular nodes
   (c) The number of radial nodes is $n - l - 1$, where *n* is the principal quantum number and *l* is the angular momentum quantum number
   (d) All orbitals of the same principal quantum number *n* have the same number of nodes.

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9. Identify the correct statement(s) from the following, which are about the radial probability density function {P(r) = $r^2R^2_{(r)}$} of H atom. {Given that $P(r) = \frac{4Z^3}{a_0^3}\, r^2\, exp^{-2Zr/a_0}$}
   (a) P(0) is zero at the nucleus
   (b) When r tends to infinity, P(r) also tends to infinity
   (c) P(r) passes through a maximum at intermediate r
   (d) The maximum of P(r) represents the most probable radius at which the electron will be found

10. The wavefunction of hydrogen atom $2s$ is given below

$$\psi_{2s} = \frac{1}{4(2\pi)^{1/2}} \left(\frac{1}{a}\right)^{3/2} \left(2 - \frac{r}{a}\right) e^{-\frac{r}{2a}}$$

    (a) Calculate the average value of r in this state.
    (b) Identify the position of the radial node.